
Solution to Exercise Sheet 4:
Tensors in probability theory
Foundations of Neuro-Symbolic AI
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Task 1:

The probability distribution $\mathbb{P}[X_0, X_1, X_2]$ is a tensor of order 3:

$$\begin{pmatrix} 0.648 & 0.015 \\ 0.072 & 0.005 \end{pmatrix} \begin{pmatrix} 0.072 & 0.135 \\ 0.008 & 0.045 \end{pmatrix} [X_0, X_1, X_2]$$

- (i) The marginal distribution of *Traffic* (X_0) and *Crash* (X_1) is the contraction

$$\mathbb{P}[X_0, X_1] = \langle \mathbb{P}[X_0, X_1, X_2] \rangle_{[X_2]} = \begin{pmatrix} 0.72 & 0.15 \\ 0.08 & 0.05 \end{pmatrix} [X_0, X_1] .$$

- (ii) The distribution of *Crash* (X_1) conditioned on *Traffic* (X_0) is the normalization

$$\begin{aligned} \langle \mathbb{P}[X_0, X_1] \rangle_{[X_0]} &= \begin{pmatrix} \frac{72}{72+15} & \frac{15}{72+15} \\ \frac{8}{8+5} & \frac{5}{8+5} \end{pmatrix} [X_0, X_1] = \begin{pmatrix} \frac{24}{13} & \frac{5}{13} \\ \frac{8}{13} & \frac{5}{13} \end{pmatrix} [X_0, X_1] \\ &\approx \begin{pmatrix} 0.828 & 0.172 \\ 0.615 & 0.385 \end{pmatrix} [X_0, X_1] . \end{aligned}$$

Task 2:

The event corresponds with the propositional formula

$$f[X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} [X_0, X_1] .$$

To compute the probability that this event holds we perform the contraction

$$\begin{aligned} \langle \mathbb{P}[X_0, X_1, X_2], f[X_0, X_1] \rangle_{[\emptyset]} &= \langle \mathbb{P}[X_0, X_1], f[X_0, X_1] \rangle_{[\emptyset]} \\ &= 0.72 + 0.15 + 0.05 \\ &= 0.92 . \end{aligned}$$

Task 3:

- (i) The variable X_1 is not independent of X_0 since

$$\mathbb{P}[X_0 = 0] = 0.97 \quad , \quad \mathbb{P}[X_1 = 0] = 0.8$$

but

$$\mathbb{P}[X_0 = 0, X_1 = 0] = 0.72 \neq 0.8 \cdot 0.97 = 0.776 .$$

- (ii) We need to investigate, whether X_0 is independent of X_2 with respect to the distributions $\mathbb{P}[X_0, X_2|X_1 = 0]$ and $\mathbb{P}[X_0, X_2|X_1 = 1]$.

a) We compute

$$\mathbb{P}[X_0, X_2 | X_1 = 0] = \frac{1}{648 + 72 + 72 + 8} \begin{pmatrix} 648 & 72 \\ 72 & 8 \end{pmatrix} [X_0, X_2] = \begin{pmatrix} 0.81 & 0.09 \\ 0.09 & 0.01 \end{pmatrix} [X_0, X_2]$$

We notice that matrix coincides with the tensor product

$$\begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix} [X_0] \otimes \begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix} [X_2] = \mathbb{P}[X_0 | X_1 = 0] \otimes \mathbb{P}[X_2 | X_1 = 0]$$

and thus, X_0 is independent of X_2 with respect to the distributions $\mathbb{P}[X_0, X_2 | X_1 = 0]$.

b) Conversely we have

$$\mathbb{P}[X_0, X_2 | X_1 = 1] = \frac{1}{15 + 5 + 135 + 45} \begin{pmatrix} 15 & 135 \\ 5 & 45 \end{pmatrix} [X_0, X_2] = \begin{pmatrix} 0.075 & 0.675 \\ 0.025 & 0.225 \end{pmatrix} [X_0, X_2]$$

We compare with the tensor product

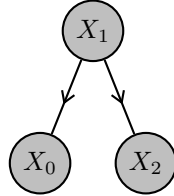
$$\mathbb{P}[X_0 | X_1 = 1] \otimes \mathbb{P}[X_2 | X_1 = 1] = \begin{pmatrix} 0.75 \\ 0.25 \end{pmatrix} [X_0] \otimes \begin{pmatrix} 0.1 \\ 0.9 \end{pmatrix} [X_2] = \begin{pmatrix} 0.075 & 0.675 \\ 0.025 & 0.225 \end{pmatrix} [X_0, X_2]$$

and notice that both are equal. Thus, X_0 is also independent of X_2 with respect to the distributions $\mathbb{P}[X_0, X_2 | X_1 = 1]$.

In total this shows that $(X_0 \perp X_2) | X_1$, thus the server traffic and the alert are independent on each other conditioned on whether the server crashes.

Task 4:

The conditional independence $(X_0 \perp X_2) | X_1$ is represented in the directed graph:



We get the Bayesian network by decoration of the directed hyperedges by

$$\begin{aligned} \mathbb{P}[X_1] &= \begin{pmatrix} 0.2 \\ 0.8 \end{pmatrix} [X_1] \\ \mathbb{P}[X_0 | X_1] &= \begin{pmatrix} \frac{24}{5} & \frac{8}{13} \\ \frac{29}{5} & \frac{13}{5} \end{pmatrix} [X_0, X_1] \\ \mathbb{P}[X_2 | X_1] &= \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix} [X_2, X_1] . \end{aligned}$$